

Loss functions

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can calculate per-example squared error and mean squared error.

I can distinguish residual cancellation from a low loss.

I can separate training loss from independent evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Calculate per-example squared error and mean squared error

Distinguish residual cancellation from a low loss

Separate training loss from independent evaluation

Retrieve what you already know

Does knowing a network output tell us whether it is accurate?

My starting explanation

This week's end goal

By the end, I can explain loss functions using a traced example and a stated limitation.

2. Notice the evidence

A loss assigns a numerical penalty to a prediction compared with a target. For this regression lesson, residual $e = \text{prediction} - \text{target}$, and squared error is e squared. Squaring makes both positive and negative residuals contribute nonnegative penalties. We use ordinary mean squared error without a one-half factor.

Example · Target · Prediction · Residual · Squared error

Example: A

Target: 1

Prediction: 2

Residual: 1

Squared error: 1

Example: B

Target: 3

Prediction: 2

Residual: -1

Squared error: 1

Example: C

Target: 2

Prediction: 4

Residual: 2

Squared error: 4

The mean squared error, or MSE, is $(1 + 1 + 4) / 3 = 2$. It summarises these three examples, not every

possible future input. If targets use cm, residuals use cm and MSE uses square cm.

List the given facts and one fact that is not supplied.

3. See how it works

For targets [1, 3] and predictions [2, 2], the mean residual is zero because 1 and -1 cancel. MSE is 1, correctly recording that both predictions miss. A mean residual can describe directional tendency, but it is not a substitute for an error magnitude.

```
def mse(predictions, targets):  
    if not targets or len(predictions) != len(targets):  
        raise ValueError("Need equal nonempty sequences")  
    errors = [(p - t) ** 2  
              for p, t in zip(predictions, targets)]  
    return sum(errors) / len(errors)  
  
print(mse([2, 2, 4], [1, 3, 2])) # 2.0  
print(mse([2, 2], [1, 3]))      # 1.0
```

The guard prevents silently ignoring an unmatched example or dividing by zero. It still assumes paired, numeric, valid records. Wrong row alignment produces a meaningless comparison even when list lengths agree.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Targets [0, 4] have predictions [2, 2]. Compare mean residual and MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Choose MAE instead of MSE as the sole criterion.

Prediction: A and B tie at MAE1.

Reason: Squared error penalises B's concentrated error more, while absolute errors both sum 2.

5. Words that unlock the idea

Word · Meaning

Word: Target

Meaning: The recorded value a model is intended to predict

Word: Residual

Meaning: Prediction minus target under the sign convention used here

Word: Squared error

Meaning: The square of a numerical residual

Word: Mean squared error

Meaning: The average squared error over the evaluated examples

Use two terms to explain a step in this week's example.

6. Follow a worked example

Use the same targets and denominator

Candidate predictions · Squared errors · MSE

Candidate predictions: P is [2, 2, 4]

Squared errors: [1, 1, 4]

MSE: 6 divided by 3 is 2

Candidate predictions: Q is [1, 2, 2]

Squared errors: [0, 1, 0]

MSE: 1 divided by 3

Candidate predictions: Baseline is [2, 2, 2]

Squared errors: [1, 1, 0]

MSE: 2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Loss functions

Compare candidate losses with the same targets and denominator.

The facts we will use

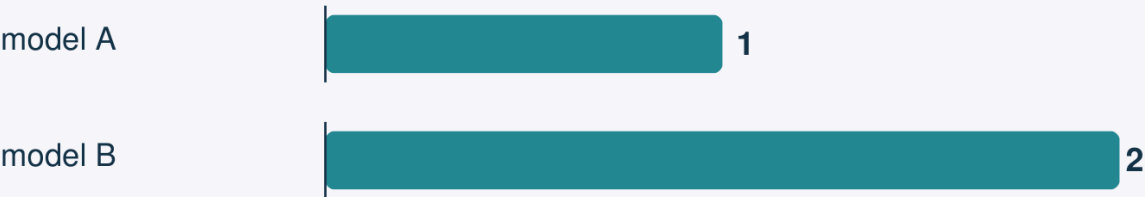
Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/8

MSE comparison on shared targets

squared target units



Bar length shows the supplied value; the marked baseline is zero.

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. A squared errors

A residuals $[-1, 1]$ give squares $[1, 1]$, MSE1.



2. B squared errors

B residuals $[0, 2]$ give squares $[0, 4]$, MSE2.



3. Compare objective

A wins under MSE; both MAEs equal1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. A squared errors

What happens and why: A residuals $[-1, 1]$ give squares $[1, 1]$, MSE1.

Step: 2. B squared errors

What happens and why: B residuals $[0, 2]$ give squares $[0, 4]$, MSE2.

Step: 3. Compare objective

What happens and why: A wins under MSE; both MAEs equal 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

For this loss calculation, the signed error is prediction minus target. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

What this example does not establish

Training loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Read the labelled picture: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Choose MAE instead of MSE as the sole criterion. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

Squaring gives larger residuals more influence: residual 4 contributes 16, four times the penalty of residual 2. This can be useful when large misses matter, but it can also make outliers influential. Mean absolute error uses absolute residuals instead; it can rank candidates differently.

MSE is appropriate here because targets are numerical quantities. A class identifier such as 0, 1 or 2 may merely name a category; its numeric gap does not automatically measure error severity. Later classification models need a loss suited to their outputs and task.

Locate the earliest unsupported step

A learner totals squared errors 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

State one repair and the checks you would rerun.

8. Practise together

For targets [2, 4] and predictions [1, 6], record residuals, squares and MSE.

Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.

Explain your prediction to a partner before comparing results.

Practise with less help: Expose cancellation

Use workbook W2 for the connected example “Loss functions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Loss functions. Proposed explanation: “Average residual is the loss”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Targets [0, 4] have predictions [2, 2]. Compare mean residual and MSE.

```
def mse(predictions, targets):  
    if not targets or len(predictions) != len(targets):  
        raise ValueError("Need equal nonempty sequences")  
    errors = [(p - t) ** 2  
              for p, t in zip(predictions, targets)]  
    return sum(errors) / len(errors)  
  
print(mse([2, 2, 4], [1, 3, 2])) # 2.0  
print(mse([2, 2], [1, 3]))      # 1.0
```

Record - Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For targets [2, 4] and predictions [1, 6], record residuals, squares and MSE.

A learner totals squared errors 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled loss functions evidence sheet to Image-pattern Network Investigation. Show how you can calculate per-example squared error and mean squared error. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

Training loss measures the cases used to adjust parameters. Validation loss can guide choices, so repeated use makes it development evidence. A reserved final test supports a separate evaluation claim only while it remains outside those choices. Always report the split, number of cases and loss definition.

A model reports MSE zero on its training rows. What has this shown, and why is it insufficient to claim perfect future predictions?

Next week studies local changes in a loss, preparing us to choose a direction for a parameter update.

A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Use the same targets and denominator

Candidate predictions · Squared errors · MSE

Candidate predictions: P is [2, 2, 4]

Squared errors: [1, 1, 4]

MSE: 6 divided by 3 is 2

Candidate predictions: Q is [1, 2, 2]

Squared errors: [0, 1, 0]

MSE: 1 divided by 3

Candidate predictions: Baseline is [2, 2, 2]

Squared errors: [1, 1, 0]

MSE: 2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Compare an unfamiliar evaluation independently

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too. State what changes in the conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too. State what changes in the conclusion.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Compare candidate losses with the same targets and denominator.

By the end, I can · How I will show it

By the end, I can: Calculate per-example squared error and mean squared error

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Distinguish residual cancellation from a low loss

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Separate training loss from independent evaluation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Calculate per-example squared error and mean squared error

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare candidate losses with the same targets and denominator.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A wins under MSE; both MAEs equal 1.

1. Compare objective
2. A squared errors
3. B squared errors

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare objective A wins under MSE; both MAEs equal 1. For this loss calculation, the signed error is prediction minus target. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Practice 2: explain the deciding evidence

Changed-case practice: Choose MAE instead of MSE as the sole criterion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. A and B tie at MAE 1.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A and B tie at MAE 1. Squared error penalises B's concentrated error more, while absolute errors both sum 2.

Repair a missing building block

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer inputs. Short

practice: What intermediate value explains the changed output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums ($a_i \cdot 5/2$). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.